

Saurabh Khanna

Department of Communication Science
Faculty of Social and Behavioural Sciences
University of Amsterdam

saurabh.khanna@uva.nl
+31 64 1684288
saurabh-khanna.github.io

APPOINTMENTS

- 2024– University of Amsterdam
Assistant Professor [tenured], Department of Communication Science
PI, Invisible Information Lab [invisible.info]
- 2024– University of Oxford
Research Fellow, Pembroke College
- 2023–24 University of Oxford
Junior Research Fellow [Nigel James Data Science Fellow], Pembroke College

EDUCATION

- Ph.D. Education Policy [Computer Science minor], Stanford University, 2023
- M.A. Economics, Stanford University, 2021
- M.A. Education, Tata Institute of Social Sciences, 2017
- B.S. Computer Science Engineering, BITS Pilani, 2012

RESEARCH AFFILIATIONS

- 2019– Public Knowledge Project, Stanford University & Simon Fraser University
- 2019– Immigration Policy Lab, Stanford University & ETH Zürich
- 2021–23 Stanford Institute for Human-Centered Artificial Intelligence
- 2021–23 Stanford Data Science
- 2021–23 Stanford GCSI Lab & World Health Organization
- 2021–23 Stanford Center for Open and Reproducible Science
- 2018–19 The World Bank
- 2015–17 CLIX [Tata Institute of Social Sciences & Massachusetts Institute of Technology initiative]

GRANTS AND AWARDS

- 2026–30 VENI: NWO Talent Programme, Dutch Research Council [EUR 382301]
- 2025–27 Fair and Resilient Societies Grant (PI), University of Amsterdam [EUR 216900]
- 2025 Responsible AI Innovation (PI), University of Amsterdam [EUR 8000]
- 2024–25 ASCoR Collaborative Grant (PI), University of Amsterdam [EUR 25000]
- 2024 Digicom Lab Seed Grant (PI), University of Amsterdam [EUR 4500]

2024	Social and Behavioural Data Science Centre Valorization Grant (PI), University of Amsterdam [EUR 1500]
2024	ARILS Methods and Interdisciplinary Research Fund (led by Jef Ausloos), University of Amsterdam [EUR 8000]
2024	Social and Behavioural Data Science Centre Interdisciplinary Grant (led by Zak Witkower), University of Amsterdam [EUR 10000]
2023	Technology Ethics & Policy Fellowship, Stanford University [USD 12000]
2021-23	Stanford Data Science Scholarship [USD 52984]
2022-23	Stanford Institute for Human-Centered Artificial Intelligence Fellowship [USD 2000]
2022-23	Digital Humanities Fellowship, Stanford Center for Spatial and Textual Analysis [USD 1500]
2018-21	William R. and Sara Hart Kimball Fellowship, Stanford University [USD 138600]
2021	Design Challenge Research Award, Stanford University and StartX [USD 9000]
2021-22	Karr Fellowship, Center for Education Policy Analysis, Stanford [USD 6000]
2021	Digital Learning Design Challenge Grant, Stanford University and StartX [USD 1500]
2020	Stanford Impact Labs Research Fellowship [USD 3500]
2020	Stanford Data Challenge Lab Member [30 Stanford students selected]
2019	Immigration Policy Lab Collaborative Research Fellow [USD 4500]
2017	Institute Silver Medal Award, Tata Institute of Social Sciences
2006-12	National Talent Search Examination Scholar, Government of India [Top 0.2% of 500000]
2008	All India Rank 562, All India Engineering Entrance Examination [Top 0.05% of 1000000]

PUBLICATIONS

Articles

2026	Cole, K., Pearman II, F. A., & Khanna, S. Big Fish or Little Pond? National Evidence on the Relationship Between Transferring and Athletic Performance Within Division I Men's College Basketball. <i>Journal of Issues in Intercollegiate Athletics</i> , 19(1), 15.
2026	Khanna, S., Chen, Z., Billedo, C., Yan, J., van Driel, I., Martelo, A. B., Sumter, S. (2026). Picturing Perceptions: An Open-Source Toolkit to Uncover Bias in Humans and Machines. arXiv preprint: 2606.13688.
2026	Khanna, S. Knowing Unknowns in an Age of Information Overload. arXiv preprint:2510.10413.
2025	Loyalka, P., Mistree, D., Fairlie, R., & Khanna, S. Job Training, English Language Skills, and Employability: Evidence from an Experiment in Urban India. <i>Economic Development and Cultural Change</i> , 73(4). doi:10.1086/734458
2024	Silverman, R. D., Darling-Hammond, E., Keane, K., & Khanna, S. (2024). The Effects of Educational Technology Interventions on Literacy in Elementary School: A Meta-Analysis. <i>Review of Educational Research</i> . doi:10.3102/00346543241261073
2023	Ulitzsch, E., Khanna, S., Rhemtulla, M., Domingue, B. A Graph-theory Based Similarity Metric Enables Comparison of Sub-population Psychometric Networks. <i>Psychological Methods</i> . doi:10.1037/met0000625
2022	Khanna, S., Ball, J., Alperin, J. P., & Willinsky, J. Recalibrating the Scope of Scholarly Publishing: A Modest Step in a Vast Decolonization Process. <i>Quantitative Science Studies</i> . doi:10.1162/qss_a_00228

- 2022 Loyalka, P., Shi, Z., Li, G., Kardanova, E., Chirikov, I., Yu, N., Hu, S., Wang, H., Ma, L., Guo, F., Liu, O. L., Bhuradia, A., Khanna, S., Li, Y., & Murray, A. The Effect of Faculty Research on Student Learning in College. *Educational Researcher*, 51(4), 265–273. doi:10.3102/0013189X221090229
- 2021 Loyalka, P., Liu, O. L., Li, G., Kardanova, E., Chirikov, I., Hu, S., Yu, N., Ma, L., Guo, F., Beteille, T., Tognatta, N., Gu, L., Ling, G., Federiakin, D., Wang, H., Khanna, S., Bhuradia, A., Shi, Z., & Li, Y. Skill Levels and Gains in College STEM Education in China, India, Russia, and the United States. *Nature Human Behavior* doi:10.1038/s41562-021-01062-3
- 2021 McFarland, D. A., Khanna, S., Domingue, B. W., & Pardos, Z. A. Education Data Science: Past, Present, Future. *AERA Open*. doi:10.1177/23328584211052055
- 2020 Silverman, R., Johnson, E., Keane, K., & Khanna, S. Beyond decoding: A meta-analysis of the effects of language comprehension interventions on K-5 students' language and literacy outcomes. *Reading Research Quarterly* doi:10.1002/rrq.346
- 2019 Loyalka, P., Liu, O. L., Li, G., Chirikov, I., Kardanova, E., Gu, L., Ling, G., Yu, N., Guo, F., Ma, L., Hu, S., Johnson, A. S., Bhuradia, A., Khanna, S., Froumin, I., Shi, J., Choudhury, P. K., Beteille, T., Marmolejo, F., & Tognatta, N. Computer science skills across China, India, Russia, and the United States. *Proceedings of the National Academy of Sciences*, 201814646 doi:10.1073/pnas.1814646116
- 2017 Khanna, S. Through the Sociological Lens: Learning Mathematics in a Mumbai Classroom. *For the Learning of Mathematics* 37(3), 24-26. jstor.org/stable/26548466

Book Chapters

- 2023 Khanna, S., & Willinsky, J. What Those Responsible for Open Infrastructure in Scholarly Communication Can Do about Possible Predatory Practices. *Predatory Practices in Scholarly Communication and Publishing: Causes, Forms, Implications, and Solutions*. London Routledge. doi:10.1590/SciELOPreprints.3474

Papers in development

[In reverse chronological order of project initiation]

- Invisible Information: Theory, Measurement, and Boosting
- Invisible Conflicts: Addressing the gap between human suffering and global media attention [NWO VENI grant]
- Invisible Languages of the LLM Universe [FRS Midsize grant]
- *heterogeneity*: R package for Heterogeneity in Media Effects using Machine Learning Approaches
- Invisible books and contextualizing open libraries
- The films we do not see
- The Evolution of Innovation in Communication Science over the last 100 years
- AI Mediated Communication with Non-Human Animals
- Impact of active learning on STEM achievement: Systematic review and new causal evidence [Under Review]
- Affirmative Action, Quality of Instruction and Lower-Caste Student-Faculty Interactions: Evidence from Randomly Assigned Classes in Engineering Colleges in India. Link. [Under Review]
- Does Diversity Reduce Segregation? Experimental Evidence from a Large-Scale, Nationally Representative, Longitudinal Study of Colleges in India.
- Automated Chat Application Surveys Using WhatsApp. *SocArXiv* <https://repec.iza.org/dp15263.pdf> [Under Review]
- Gender Gap in Refugee Integration.
- A Revolution in Information Warfare: Throttling, Deamplification, and Deplatforming.

- The Effect of Sleep on Learning: A Longitudinal Study of Undergraduates in China, India, and Russia.
- English Language Skills Reduce Poverty [In Press]

Conference Proceedings

[*current till 12.2024]

- 2024 Giebel, S., Khanna, S., Marbach, L., Soysal, Y. Defining Diversity Discourse in Global Higher Education: Computationally Comparing India, the UK, and Germany. *American Sociological Association Annual Conference, Montreal*
- 2024 Giebel, S., Khanna, S., Marbach, L., Soysal, Y. Defining Diversity Discourse in Global Higher Education: Computationally Comparing India, the UK, and Germany. *16th European Sociological Association Conference, Porto*
- 2024 Khanna, S. Incomplete Information. Theory, Measurement, and Boosting. *European Communication Science Event, Nijmegen*
- 2023 Khanna, S. Measuring Information Completeness. *International Conference on Statistics and Data Science, Lisbon*
- 2023 Khanna, S. Recalibrating the Scope of Scholarly Publishing: A Modest Step in a Vast Decolonization Process. *52nd LIBER Annual Conference. Open and Trusted: Reassessing Research Library Values, Budapest*
- 2023 Khanna, S. Knowing Unknowns in an Age of Incomplete Information. *iConference 2023. Normality, Virtuality, Inclusivity, Physicality, Barcelona*
- 2022 Khanna, S. Invisible Information in Ranked Contexts. *ACM Conference on Fairness, Accountability, and Transparency (FAccT), Seoul*
- 2022 Fairlie, R., Khanna, S., Loyalka, P., Sachdeva, G. Affirmative Action, Quality of Instruction and Lower-Caste Student-Faculty Interactions: Evidence from Randomly Assigned Classes in Engineering Colleges in India. *IGIDR Second Biennial Conference on Development, Mumbai*
- 2022 Khanna, S. How Much Do We Not Know? Estimation and Implications. *Stanford Data Science Conference*
- 2022 Khanna, S. Visibility Curves & The Internet. *iConference 2022. Information for a Better World: Shaping the Global Future*
- 2021 Khanna, S. Invisible Knowledge & The Internet: Estimation and Implications. *Stanford HAI & ETH AI Data-Centric AI Workshop*
- 2022 Khanna, S. Knowing Unknowns in an Age of Information Overload. *OpenSciComm, Belgrade*
- 2021 Khanna, S., Raoni, J., Smecher, A., Alperin, J. P., Ball, J. Location of known journals using PKP's Open Journal Systems. *Open Publishing Fest*
- 2020 Khanna, S. Is the Internet Impartial? Diversity in Knowledge Recommendation in Online Networks. *2020 Sunbelt Conference, Paris*
- 2020 Khanna, S., Loyalka, P. Friendship Networks in College. *Conference on Educational Data Science, Stanford*
- 2020 Khanna, S., Loyalka, P. Affirmative Action and Social Integration in College. *SREE Spring 2020 Conference, Practical Significance and Meaningful Effects: Learning and Communicating What Matters, Arlington*
- 2017 Khanna, S., Gajinkar, A., Roy, A., Chatterji, A., Bapat, A., Bose, A. Rigor as Familiarity in Mathematics Assessments. *41st Annual Meeting of the International Group for the Psychology of Mathematics Education, Singapore*

- 2017 Bapat, A., Khanna, S., Srinivas, S., Thirumalai, B., Kumar, R., Rahaman, J., Chougale, S., Bose, A. Facilitating Geometry Learning through Blended Curriculum. *41st Annual Meeting of the International Group for the Psychology of Mathematics Education, Singapore*
- 2016 Srinivas, S., Khanna, S., Rahaman, J., & Kumar, V. Designing a Game-Based Learning Environment to Foster Geometric Thinking. *2016 IEEE Eighth International Conference on Technology for Education (T4E)* (pp. 72-79). IEEE <https://ieeexplore.ieee.org/document/7814798>
- 2016 Khanna, S. Cost-effective pedagogies for implementing efficient inclusive schooling. *6th International Conference on Science of Human Learning, Delhi*
- 2016 Khanna, S. Designing evidence based games to assess mathematical thinking. *TSG 42 of the 13th International Congress on Mathematics Education, Hamburg*

Datasets

- 2024 Khanna, S., Raoni, J., Smecher, A., Alperin, J. P., Ball, J., Willinsky, J. Location of known journals using PKP's Open Journal Systems V3. doi:10.7910/DVN/OCZNVY
- 2021 Khanna, S., Raoni, J., Smecher, A., Alperin, J. P., Ball, J. Location of known journals using PKP's Open Journal Systems V1. *Open Publishing Fest*. doi:10.7910/DVN/OCZNVY

Media Coverage

- 2024 Global Biases in Research Metrics and Fairness in Knowledge Propagation. Knowledge Equity Network
- 2023 Beyond Web of Science and Scopus there is already an open bibliodiverse world of research – We ignore it at our peril. London School of Economics Impact
- 2021 Many science courses 'fail to improve critical thinking'. Times Higher Education
- 2021 Supertest evaluates performance of engineering students in Russia, U.S., India, China. EurakAlert
- 2021 Computer science college seniors in U.S. outperform peers in China, India and Russia, new research says. phys.org
- 2019 Who Trains the Best Computer Scientists? World Bank Blogs
- 2019 U.S. students excel in computer science, outscoring China, India and Russia, Stanford study says. Mercury News
- 2019 U.S. College Seniors Beat Peers from China, India, Russia on Computer Science Exam. Campus Technology
- 2019 Study shows US computer science graduates have the edge. University World News

INVITED TALKS

- 2024 Global Biases in Research Metrics and Fairness in Knowledge Propagation. Knowledge Equity Network, University of Leeds
- 2024 Generative AI and Communication. Digital Communications Lab, University of Amsterdam
- 2024 Knowing Unknowns. Social and Behavioural Data Science Centre, University of Amsterdam
- 2024 Invisible Information: Theory, Measurement, and Boosting. AI, Media and Democracy Lab
- 2024 PictoPercept: Innovating bias assessment through pairwise visual wikisurveys. Digital Communications Lab, University of Amsterdam
- 2023 Misinfotainment: The blurring lines between evidence and entertainment on the Internet. Pembroke College, University of Oxford

2023	Decoding Data: Turning Numbers into Knowledge. Pembroke College, University of Oxford
2023	Computer Science for Social Good. Pembroke College, University of Oxford
2023	DOAJ at 20: Global
2023	Forum for Open Research in MENA (FORM)
2023	Public Knowledge Project Annual General Meeting
2022	Leveraging Git for Research. Stanford Graduate School of Education
2021	Invisible Knowledge and the Internet: Estimation and Implications. Leverhulme Centre for Demographic Science, University of Oxford
2021	Assessing Fairness in Internet Search. Stanford Social Entrepreneurial Students' Association
2020	Open-source tools for analyzing educational survey data. All India Council for Technical Education, Ministry of Human Resource Development (MHRD), Government of India
2017	Reservation under the Right to Education Act. Quality Education Support Trust
2015	Identity and the Textbook. Center for Education, Innovation and Action Research

COURSES TAUGHT

University of Amsterdam

Computational Communication Science

Digital Analytics

Always Connected: Insights in Youth, Media, and Technology

Entertainment Communication

Stanford University

EDUC 306Y: Education and Economic Development

SOC 302B: Education Data Science II [Teaching Assistance]

EDUC 306A: Economics of Education in the Global Economy [Teaching Assistance]

EDUC 200A: Introduction to Data Analysis and Interpretation [Teaching Assistance]

Open access resources created

The (In)visible Hub

PictoPercept

Leveraging Git for Research. Stanford Graduate School of Education [content]

Assessing Fairness in Internet Search. Stanford Social Entrepreneurial Students' Association

Open-source tools for analyzing educational survey data. All India Council for Technical Education, Ministry of Human Resource Development, Government of India [content]

SERVICE

Academic Peer Review

Journal of Social Computing (Guest Editor)

Quantitative Science Studies

Journal of Open Source Software

Educational Researcher

AERA Open

ICA (International Communication Association) Computational Methods Division

FAccT (ACM Fairness, Accountability, and Transparency)

AISTATS (Society for Artificial Intelligence and Statistics)

SREE (Society for Research on Educational Effectiveness)

iConference

University

TLC FMG AI Team, University of Amsterdam

Advisory Board Member, Digital Communications Lab, University of Amsterdam

Advising & Mentorship

Jade van Dorth, PhD in Communication Science, University of Amsterdam

Jinyi Yang, MSc in Communication Science, University of Amsterdam

Zhijun Chen, MSc in Communication Science, University of Amsterdam

Xinkangrui Gao, MSc in Communication Science, University of Amsterdam

Elean Huang, MSc in Communication Science, University of Amsterdam

Chia-Ai Hsu, MSc in Communication Science, University of Amsterdam

Bohao Hu, MSc in Communication Science, University of Amsterdam

Ruslan Pinski, MSc in Communication Science, University of Amsterdam

Hansol Lee, PhD in Education Data Science, Stanford University

Jodi Anderson Jr., MA in Policy Organization and Leadership Studies, Stanford University

Aparajitha Suresh, MA in International Education Policy Analysis, Stanford University

Vehbi Tandogan, MA in International and Comparative Education, Stanford University

Meng Guo, MA in International Education Policy Analysis, Stanford University

Pok Jing Ho, MA in International and Comparative Education, Stanford University

INDUSTRY EXPERIENCE

2012-13 Software Engineer, Juniper Networks, Bengaluru

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